

VPN Troubleshooting Runbook

Purpose: diagnose remote-access VPN failures without weakening security controls. This runbook applies to a company-managed device using an approved VPN client. Names of gateways, profiles, and support contacts must come from the employee's approved company portal.

Rapid triage

- Record the exact error, time, device name, operating system, VPN client version, and network type.
- Confirm ordinary internet access by opening two trusted public sites. If neither opens, troubleshoot the local network before the VPN.
- Confirm date, time, and time zone are correct. Large clock differences can invalidate certificates and tokens.
- Check the approved status page or service notice. Do not repeatedly reconnect during a known outage.
- Determine scope: one user, one device, one network, or multiple employees. Scope changes the likely cause.

Security stop conditions

- Never share passwords, MFA codes, recovery codes, certificates, private keys, or session tokens.
- Deny and report an MFA prompt that the employee did not initiate.
- Stop and contact IT for certificate warnings, an unrecognized VPN profile, suspected account compromise, or instructions to disable firewall or endpoint protection.

VPN Connection and Authentication Failures

Cannot reach the VPN gateway

Common messages: "VPN server unavailable," "Gateway not responding," "The network connection could not be established," or Windows error 809.

- Verify public internet access and then close and reopen the approved VPN client.
- Switch once between wired and trusted Wi-Fi, if available. Success on one network suggests local router, captive portal, or ISP filtering.
- Complete any hotel, airport, or guest-network captive portal in a browser before starting VPN. Avoid sensitive work on untrusted networks.
- Restart the laptop. Do not change router port-forwarding, registry, firewall, DNS, or VPN protocol settings unless IT provides an approved procedure.
- If several employees fail at the same time, treat it as a possible service incident and escalate rather than reinstalling clients.

Credentials rejected or sign-in loops

Common messages: "Authentication failed," "Credentials rejected," "Account locked," error 691, or repeated sign-in prompts.

- Confirm the user can sign in to the approved company identity portal. Never test a password on an unfamiliar site.
- After a recent password change, fully exit the VPN client and sign in with the new password. Reboot if old credentials remain cached.
- Stop after a small number of careful attempts. Repeated failures can extend an account lockout.
- Approve only an MFA prompt initiated by the current VPN sign-in. Check that the authenticator device has internet access and correct time.
- If web sign-in succeeds but VPN authentication fails, capture the VPN error and contact IT; VPN access, group membership, device compliance, or profile assignment may require administrator review.

Certificate or compliance failure

Messages may mention certificate expired, certificate not trusted, device not compliant, conditional access, or access denied by policy.

- Confirm system time and install pending approved operating-system updates if permitted.
- Connect the device to the internet long enough for management policy to synchronize, then retry once.
- Do not bypass certificate validation, install a certificate sent through chat, or remove device-management software.
- Escalate with the certificate subject/expiry date or compliance message, but never export a private key.

VPN Connected but Work Resources Fail

Tunnel connects but internal sites do not open

- Verify the VPN client explicitly shows Connected and note the assigned connection time.
- Try two known internal resources. One failing resource suggests that service; all internal resources failing suggests routing, DNS, authorization, or tunnel policy.
- Use the full approved internal hostname rather than a saved shortcut. Do not guess internal server addresses.
- Disconnect once, wait 30 seconds, and reconnect. If failure persists, capture the internal hostname, browser message, and whether public sites still work.
- Do not run copied network-reset scripts or change DNS servers without IT approval.

File share, remote desktop, or application unavailable

- Confirm the specific application is approved for remote access and that the target system is expected to be online.
- Record whether the message is timeout, name not found, access denied, or credentials rejected; each points to a different layer.
- An access-denied response after the VPN connects is normally an authorization issue, not proof that the tunnel failed.
- For remote desktop, confirm the destination name came from an approved source. Do not expose remote desktop directly to the internet.

Slow VPN or frequent disconnection

- Compare internet stability without VPN. Packet loss, weak Wi-Fi, roaming between access points, or laptop sleep can drop the tunnel.
- Move closer to the access point, pause large downloads and cloud synchronization, and keep the laptop awake during one test.
- If permitted, test a wired connection. Record the approximate disconnect interval and whether calls, all traffic, or only one application is affected.
- Do not use consumer "VPN optimizer" software or an unapproved personal VPN at the same time.

VPN Verification and Escalation

Resolution checks

- The client remains Connected for at least ten minutes on a stable network.
- The employee can open an approved internal resource and complete the original task.
- No certificate warning, unexpected MFA prompt, or repeated credential request appears.
- Disconnect and reconnect once to confirm the resolution persists.

Contact IT immediately

- Unexpected MFA prompts, suspected credential theft, an unknown profile, or a certificate/private-key warning.
- Multiple employees are affected or a business-critical service is unavailable.
- The client reports account disabled, device blocked, compliance failure, or access denied by policy.
- The VPN causes a crash, blue screen, or loss of network connectivity after reboot.

Standard ticket evidence

- Employee contact, device name/asset tag, operating system, VPN client version, and connection type.
- Exact error text/code and timestamp with time zone.
- Whether public internet works, whether identity-portal sign-in works, and whether other employees are affected.
- Internal resource names affected and safe steps already attempted.
- A sanitized screenshot with passwords, tokens, email content, and confidential data removed.

Do not attach VPN diagnostic bundles unless IT requests them through an approved secure channel; logs can contain internal addresses and identifiers.